Microeconomics Group Assignment 4

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Q50

Q50

$$C(q) = q^2 + 16$$

$$MC = 2q$$

$$AC = \frac{q^2 + 16}{q} = q + \frac{16}{q}$$

$$Q^* = 2q = q + \frac{16}{q}$$

$$q = \frac{16}{q}$$

$$q^2 = 16 \Rightarrow \sqrt{q^2} = \sqrt{16}$$

$$q = 4$$

Q53

53) $C^{o}_{SR}(q,\bar{K}) = \underbrace{w\cdot L^{*}_{SR}(q,\bar{K})}_{a}$

$$C^{o}_{LR}(q) = \underbrace{w\cdot L^{*}_{LR}(q)}_{(a)} + \underbrace{r\cdot K^{*}_{LR}(q)}_{(b)}$$

question; is $C^{o}_{LR}(q)$ always $\leq C^{o}_{SR}(q,\bar{K})$

$\boxed{\text{given same}(q)}$

False. If the shortrun opportunity cost (a) is the same in the Long run, the long-run opportunity cost $(a+b)$ will be larger than the shortrun (a). This scenario will occur when the level of capital (K) is equal in the SR as in the LR $(\bar{K}=K^{*}_{LR})$

in the long run, capital and Labour costs and units can vary making it uncertain to determain if the costs will be larger or smaller as compared to the SR.

Q65

65

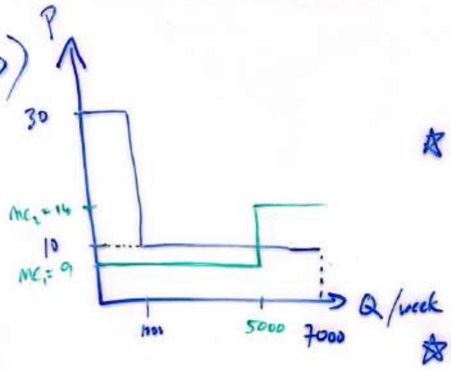
a) Firms will produce when $MR \geq MC$

$MR = 10$, $MC_{q=5k} = 9$, $MC_{q=6,7k} = 14$

Therefore, $q = 5000$ generic case //

because when $q > 5000$, $MC = 14$

65b)



$C(q) = F + V(q)$

☆ First 1000 cases (branded)

$\pi_B = 30(1000) - [1000 + 5 \times 1000 + 4 \times 1000]$

$= 20K$

☆ Firm will produce up to $MR \geq MC$

at $\boxed{q = 5000}$, $MC_2 = 14$, exceeds $MR = 10$

therefore, next 4000 case,

$\pi_G = 10(4000) - [5 \times 4000 + 4 \times 4000]$

$= 40k - 36k$

$= 4k$

Firm will produce $q_B = 1000$ & $q_G = 4000$